A

PROJECT REPORT

ON

MINI PROJECT

ENTITLED

"*Tall Tax Management System*"

Fundametals of IOT(4360703)

SUBMITTED BY

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To

COMPUTER ENGINEERING DEPARTMENT

C U SHAH GOVT. POLYTECHNIC – SURENDRANAGAR

GUJARAT TECHNOLOGICAL UNIVERSITY – AHMEDABAD

MAY – 2025

C.U.SHAH GOVERNMENT POLYTECHNIC - WADHWAN

COMPUTER ENGINEERING

CERTIFICATE

This is to certify that the Mini Project entitled **“Tall Tax Management System”** has been carried out by **CHANPURA KEYUR D.** under my guidance in partial fulfillment of the degree of Diploma Engineering in **COMPUTER ENGINEERING** 6th Semester of Gujarat Technological University, Ahmedabad during the academic year 2025.

Guides:

Mr. J.N.SOLANKI	Mr. K.G.Patel
Internal Guide	Head of Department

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Introduction to Project

- Automation is rapidly transforming various industries, making everyday tasks more efficient and convenient. One such application is the automation of toll tax systems, where technology replaces manual operations to improve traffic flow and reduce human error. In this project, we have designed a simple yet effective Automated Toll Tax System using Arduino Uno, an Ultrasonic Sensor, and a Servo Motor.
- This system works by detecting the presence of a vehicle using the ultrasonic sensor. When a vehicle is detected within a specified range, the Arduino sends a signal to the servo motor, which automatically opens the toll barrier, allowing the vehicle to pass without human intervention. After a short delay, the barrier closes, ready for the next vehicle.
- The project uses easily available components and straightforward coding, making it ideal for beginners and students looking for an interesting mini-project for a science fair or a class assignment. It introduces fundamental concepts of automation, sensors, and actuators while promoting the understanding of real-world applications of embedded systems.
- Through this report, step-by-step instructions, circuit diagrams, and a detailed explanation of the code are provided to guide anyone interested in replicating or improving the project.

Working of Project

- Vehicle Detection:
  The Ultrasonic Sensor (HC-SR04) continuously monitors the distance in front of the toll gate. It sends ultrasonic waves and measures the time it takes for the waves to bounce back from an object (vehicle).

- Signal to Arduino:
  When a vehicle is detected within a pre-defined range (for example, less than 20 cm), the ultrasonic sensor sends a signal to the Arduino Uno.

- Opening the Gate:
  Upon receiving the signal, Arduino controls the Servo Motor to rotate to a specific angle (usually 90 degrees), which lifts the toll barrier, allowing the vehicle to pass through.

- Delay and Closing the Gate:
  After a short delay (for example, 5 seconds), which gives the vehicle enough time to pass, the Arduino sends a command to the servo motor to rotate back to its original position (0 degrees), thus closing the gate.

- System Reset:
  The system resets and is ready for the next vehicle. The ultrasonic sensor again starts measuring the distance to detect the next approaching vehicle.

Advantages

1. Automation.
2. Cost-Effective.
3. Time-Saving.
4. Energy Efficient.
5. Beginner-Friendly.
6. Real-World Application.

Disadvantages

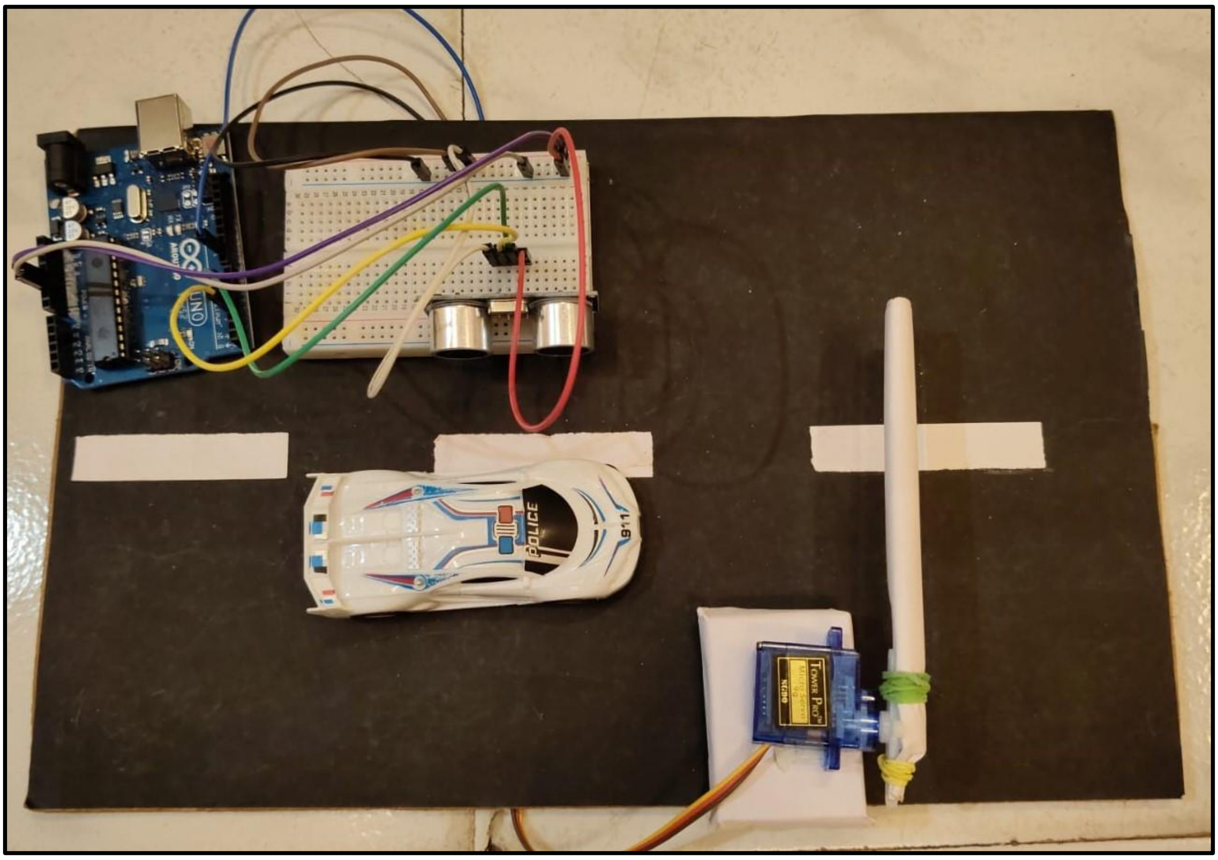
1. Limited Range.
2. Weather Sensitivity.
3. Security Issues.
4. Mechanical Wear and Tear.
5. Single Vehicle Detection.

Code of Project

```
#include<Servo.h>

Servo myservo;

const int trigPin=3;

const int echoPin=5;

long tmeduration;

int distance;


void setup() {

myservo.attach(9);

pinMode(trigPin,OUTPUT);

pinMode(echoPin,INPUT);

Serial.begin(9600);

}

void loop() {

digitalWrite(trigPin,LOW);

delayMicroseconds(2);

digitalWrite(trigPin,HIGH);

delayMicroseconds(10);

digitalWrite(trigPin,LOW);
```

```
tmeduration=pulseIn(echoPin,HIGH);
distance=(0.034*tmeduration)/2;


if(distance<=10){


myservo.write(90);
  }
else{
myservo.write(0);}


Serial.print("distance:");
Serial.println(distance);


delay(1);


}
```

Screenshots



Application

1. Automated Toll Booths.
2. Parking Management Systems.
3. Security Checkpoints.
4. Smart City Projects.
5. Educational Purposes.
6. Event Management.

Summary

- In this project, we have successfully designed and implemented an Automated Toll Tax System using Arduino Uno, an Ultrasonic Sensor, and a Servo Motor. The system detects the presence of a vehicle using the ultrasonic sensor and automatically opens the gate by rotating the servo motor. After a short delay, the gate closes again, ready for the next vehicle.
- This simple and cost-effective project helps reduce human effort, saves time, and improves traffic flow at toll booths. It demonstrates the basic principles of automation, sensor technology, and embedded systems, making it a perfect project for beginners, science fairs, and educational purposes.
- Although this model has some limitations, such as basic detection and limited security features, it provides a strong foundation for developing more advanced systems with features like payment integration, vehicle recognition, and remote monitoring.
- Overall, this project highlights how automation can play a crucial role in modern transportation systems and offers many opportunities for future improvements and applications.